Here are the complete lecture notes for "R-squared, Clearly Explained!!!".

You can copy everything below and paste it directly into a new "Text" cell in Google Colab. The math will render perfectly.

Lecture Notes: R-squared, Clearly Explained

I. Introduction to R-squared (R^2)

R-squared (or R^2) is a metric of correlation that is easy to compute and intuitive to interpret.

We are often already familiar with the standard metric for correlation, "plain old R."

- **For R, values close to 1 or -1 indicate a *strong* relationship (e.g., weight and size).**
- **For R, values close to 0 indicate a *weak* relationship (lame).**

Why do we need R^2 if we already have R?

The primary advantage of R^2 is its ease of interpretation.

- **Example: Is a correlation of $R = 0.7$ "twice as good" as $R = 0.5$? It's not obvious.**
- **In contrast, R^2 values are directly proportional. An $R^2 = 0.7$ is literally 1.4 times as good as an $R^2 = 0.5$.**
- **R^2 is also very intuitive to calculate.**

II. The Core Concept: Variation

To understand R^2, we must first understand variation.

Step 1: The Baseline (Variation around the Mean)

Let's imagine a dataset where we plot mouse weight on the Y-axis.

1. **First, we calculate the mean (average) of all the mouse weights. We can plot this as a flat, horizontal line across the graph.**
2. **Next, we measure how "spread out" the data is from this mean. This is the Total Variation.**
3. **We calculate this variation as the Sum of the Squared Differences between each data point (y_i) and the mean ($\bar{y}$).**

The formula for this Total Variation, which we can call the Sum of Squares Total (SS_{total}), is:

$$SS_{total} = \sum_i (y_i - \bar{y})^2$$

- **Note: We *square* the differences so that the points below the mean (which have a negative difference) don't cancel out the points above the mean (which have a positive difference).**

Step 2: Adding a Predictive Model (The Regression Line)

Now, let's introduce a new variable, Mouse Size, on the X-axis.

- **The data points are the same, they are just *reordered* by size.**
- **Because the Y-values haven't changed, the Mean Line and the Total Variation (SS_{total}) are *exactly the same as before*.**

Key Question: Given that we know a mouse's *size*, is the mean line still the best way to predict its *weight*?

Answer: No. We can do much better by fitting a regression line (a sloped line) to the data. This new line allows us to make a much better prediction.

Step 3: Comparing the Models

How much *better* does our new regression line fit the data compared to the original mean line? This is what R^2 measures.

To quantify this, we need a new metric: the variation *around the new line*.

1. **This is also a Sum of Squared Differences, but this time it's the difference between each actual data point (y_i) and the *predicted value on the regression line* ($\hat{y}_i$).**
2. **We can call this the Sum of Squares Residual ($SS_{residual}$). "Residual" means "left-over" variation that the line *doesn't* explain.**

The formula for this Residual Variation is:

$$SS_{residual} = \sum_i (y_i - \hat{y}_i)^2$$

III. The Formula for R-squared (R^2)

R^2 is the percentage of variation that is *explained* by the new line.

We calculate it by finding out how much variation our new line *removed*.

- **Total Variation: SS_{total} (Variation around the mean)**
- **Unexplained Variation: $SS_{residual}$ (Variation left over around the line)**
- **Explained Variation: $SS_{total} - SS_{residual}$**

We express this as a percentage of the total:

$$R^2 = \frac{SS_{total} - SS_{residual}}{SS_{total}}$$

Or, written with the summation formulas:

$$R^2 = \frac{\sum(y_i - \bar{y})^2 - \sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

Properties of R^2:

- **It always ranges from 0 to 1 (or 0% to 100%).**
- **It's a percentage.**
- **R^2 can never be greater than 1, because the variation around the best-fit line ($SS_{residual}$) can never be greater than the variation around the mean line (SS_{total}).**

IV. Worked Examples

Example 1: A Good Fit (Mouse Size vs. Weight)

Let's plug in some numbers for our mouse size vs. weight plot.

- **Variation around the mean (SS_{total}) = 32**
- **Variation around the line ($SS_{residual}$) = 6**

Calculation:

$$R^2 = \frac{32 - 6}{32}$$
$$R^2 = \frac{26}{32}$$
$$R^2 = 0.81 \text{ or } 81\%$$

Interpretation:

- **This means there is 81% less variation around the regression line than there was around the mean line.**
- **Key Takeaway: The "size-weight relationship" *accounts for 81%* of the total variation in weight.**

Example 2: A Bad Fit (Sniffing a Rock vs. Weight)

Now, let's compare weight to a useless predictor: "time spent sniffing a rock".

- **Variation around the mean (SS_{total}) = 32 (this never changes for the same Y-variable)**
- **Variation around the line ($SS_{residual}$) = 30**

Calculation:

$$R^2 = \frac{32 - 30}{32}$$

$$R^2 = \frac{2}{32}$$
$$R^2 = 0.06 \text{ or } 6\%$$

Interpretation:

- **This means there is only 6% less variation around the regression line.**
- **Key Takeaway: The "sniff-weight relationship" only *accounts for 6%* of the total variation. This is a very poor model.**

V. How to Interpret R and R^2 in Practice

The R^2 Mindset

- **If someone says $R^2 = 0.9$ (or 90%):**
  - **You think: "Very good! The relationship between these two variables explains 90% of the variation in the data."**
- **If someone says $R^2 = 0.01$ (or 1%):**
  - **You think: "Dag! Who cares if this is 'statistically significant'? It only accounts for 1% of the variation. Something *else* must be explaining the other 99%."**

The R vs. R^2 Mindset

R^2 **is simply the square of** R**.**

$$R^2 = (R)^2$$

This is why R^2 is much more intuitive.

- **If someone says $R = 0.9$:**
  - **You think: "$0.9 \times 0.9 = 0.81$". This relationship explains 81% of the variation.**
- **If someone says $R = 0.5$:**
  - **You think: "$0.5 \times 0.5 = 0.25$". This relationship explains 25% of the variation.**

This solves our original problem:

- **"How much better is $R = 0.7$ than $R = 0.5$?"**
- **Answer:**
  - $R_1 = 0.7 \rightarrow R_1^2 \approx 0.49$ **(Let's call it 50%)**
  - $R_2 = 0.5 \rightarrow R_2^2 = 0.25$
- **Conclusion: The first correlation is twice as good as the second, because it explains 50% of the variation, while the second only explains 25%. This is *not***

obvious from the R-values (0.7 vs 0.5), but *is* crystal clear from the R^2 values (50% vs 25%).

The One Downside of R^2

- **R^2 does not indicate the direction of the correlation (positive or negative).**
- **Squaring a negative number (e.g., $R = -0.8$) results in a positive R^2 (e.g., $R^2 = 0.64$).**
- **Solution: You must state the direction if it's not obvious, e.g., "The variables were *negatively correlated* with $R^2 = 0.64$."**

VI. Final Summary

1. **R^2 is the percentage of variation in the Y-variable that is explained by the relationship with the X-variable.**
2. **If someone gives you a value for "plain old R," square it in your head to understand the *true* strength and importance of the relationship.**

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Start coding or generate with AI.
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